

Rigid-body contact with length-scaled velocity and momentum

Normal and tangential restitution, friction, and experimental comparisons

Viktor's notation: $\ell\boldsymbol{\omega}$ and $\mathbf{L}/\ell$

6 October 2026

Model fidelity correction (6 October 2026). The historical force-only predictions and comparisons in this draft do not validate the requested motion-directed linear and angular impulse model. Its direction t follows relative contact velocity, and s follows relative angular velocity; these are not an arbitrary tangent basis. The general wrench identities below do not supply that constitutive law. The implementation and comparisons require revision against the original directional equations.

Abstract

A fixed reference length places linear and angular motion in one dimensionally consistent vector. Its dual places linear and angular momentum in one vector without changing power, kinetic energy, contact response, or physical predictions. We derive shared-point contact, simultaneous impulses, normal and tangential restitution, and a Coulomb capacity constraint. Public rubber-ball and limestone experiments provide partial agreement, not independent validation of all material parameters. Geometric uncertainty and a remaining rubber spin discrepancy are retained explicitly.

1 The reference length and the dual variables

Let $\ell > 0$ be a fixed reference length, $\mathbf{v}$ the centre-of-mass velocity, $\boldsymbol{\omega}$ the angular velocity, $\mathbf{p} = m\mathbf{v}$ the linear momentum, and $\mathbf{L} = \mathbf{I}\boldsymbol{\omega}$ the angular momentum about the centre of mass. All spatial vectors and the inertia tensor use the same world frame. Define

$$\mathbf{V} = \begin{bmatrix} \mathbf{v} \\ \ell\boldsymbol{\omega} \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} \mathbf{p} \\ \mathbf{L}/\ell \end{bmatrix}, \quad \mathbf{M} = \begin{bmatrix} m\mathbf{1} & 0 \\ 0 & \mathbf{I}/\ell^2 \end{bmatrix}. \quad (1)$$

Then $\mathbf{P} = \mathbf{M}\mathbf{V}$. Every component of $\mathbf{V}$ has units m/s, every component of $\mathbf{P}$ has units kg m/s, and every block of $\mathbf{M}$ has units kg. The dual pairing is unchanged:

$$\mathbf{V}^T \mathbf{P} = \mathbf{v}^T \mathbf{p} + \boldsymbol{\omega}^T \mathbf{L}, \quad T = \frac{1}{2} \mathbf{V}^T \mathbf{M} \mathbf{V}. \quad (2)$$

For force $\mathbf{f}$ and centre-of-mass torque $\boldsymbol{\tau}$, the corresponding scaled wrench is $\mathbf{F} = [\mathbf{f}^T, \boldsymbol{\tau}^T/\ell]^T$ and power is $\mathbf{V}^T \mathbf{F}$. This is a coordinate choice, *not* a friction, restitution, or material-fit parameter. A different fixed length per body is allowed; use that body's length in each block. Changing the length during an integration step requires additional coordinate-derivative terms and is excluded here.

2 Relation to established spatial notation

Six-component vectors combining translation and rotation, and their dual force/momentum vectors, are established spatial-vector and screw-theory constructions. Featherstone describes

the different transformation rules for motion and force vectors; Modern Robotics derives their power pairing[5, 6]. Characteristic-length normalization also has prior use in robotics to make mixed translational/rotational Jacobians dimensionally homogeneous[7].

A particularly close precedent is Vose, Umbanhowar and Lynch (2011): their appendix footnote 1 defines a planar velocity twist with angular velocity multiplied by radius of gyration and a force wrench with torque divided by that radius[8]. Zhang, Su and Liao (2014), section 3.3, equation 21, likewise scale rotational deformation by a characteristic length and applied moment by its reciprocal[9]. These are explicit dual length scalings, rather than merely six-component notation. For fixed length, the momentum scaling follows by integrating the scaled wrench over an impulse. We have not established priority for applying this pairing to momentum.

Our choices are translation first, the fixed scale ℓ , the dual angular momentum $\mathbf{L}/\ell$, and the signed lever operator that keeps the contact-velocity formula additive. To see precisely what is being changed, define the unscaled centre-of-mass vectors and scale matrix

$$\mathbf{w} = \begin{bmatrix} \mathbf{v} \\ \boldsymbol{\omega} \end{bmatrix}, \quad \mathbf{h} = \begin{bmatrix} \mathbf{p} \\ \mathbf{L} \end{bmatrix}, \quad \mathbf{D} = \text{diag}(\mathbf{1}_3, \ell \mathbf{1}_3), \quad \mathbf{M}_0 = \text{diag}(m \mathbf{1}_3, \mathbf{I}). \quad (3)$$

Then the entire formulation follows from an invertible coordinate change and its dual:

$$\mathbf{V} = \mathbf{D}\mathbf{w}, \quad \mathbf{P} = \mathbf{D}^{-T}\mathbf{h}, \quad \mathbf{M} = \mathbf{D}^{-T}\mathbf{M}_0\mathbf{D}^{-1}, \quad \mathbf{V}^T\mathbf{P} = \mathbf{w}^T\mathbf{h}. \quad (4)$$

For the unscaled contact map $\mathbf{C}_0 = [\mathbf{1}_3 \ -\mathbf{r}_\wedge]$, the scaled map is $\mathbf{C} = \mathbf{C}_0\mathbf{D}^{-1}$. Consequently,

$$\mathbf{C}\mathbf{M}^{-1}\mathbf{C}^T = \mathbf{C}_0\mathbf{M}_0^{-1}\mathbf{C}_0^T. \quad (5)$$

The dual momentum scaling is therefore required by the chosen velocity scaling if work and energy are to be retained. The contact response is unchanged. Keeping plus signs follows from cross-product antisymmetry; omitting redundant parentheses follows from matrix associativity. Neither operation is a new algebraic rule.

The contribution of this presentation is to carry these choices consistently through 2D/3D contact, restitution, friction capacity and efficient matrix evaluation. We do not claim a new physical law, new spatial-vector mathematics, or priority for this exact notation. The sources establish substantial prior art; the present review is not an exhaustive priority search.

3 Contact velocity with plus signs

Write $\mathbf{a}_\wedge \mathbf{b} = \mathbf{a} \times \mathbf{b}$. The skew matrix obeys $\mathbf{a}_\wedge^T = -\mathbf{a}_\wedge$. For the physical contact lever arm $\mathbf{r} = \mathbf{x}_c - \mathbf{x}_{\text{COM}}$, set $\boldsymbol{\rho} = \mathbf{r}/\ell$ and define the *signed lever operator*

$$\mathbf{R} = -\boldsymbol{\rho}_\wedge. \quad (6)$$

The physical velocity at the contact is

$$\mathbf{v}_c = \mathbf{v} + \boldsymbol{\omega} \times \mathbf{r} = \mathbf{v} + \mathbf{R}\ell\boldsymbol{\omega} = \mathbf{C}\mathbf{V}, \quad \mathbf{C} = [\mathbf{1} \ \mathbf{R}]. \quad (7)$$

The plus sign is obtained by reversing the cross-product order, since $\boldsymbol{\omega} \times \mathbf{r} = (-\mathbf{r}) \times \boldsymbol{\omega}$. It does *not* mean ordinary matrix factors commute. Associativity allows $\mathbf{R}\ell\boldsymbol{\omega}$ to be written without redundant parentheses, but does not allow $\mathbf{A}\mathbf{B}$ to be replaced by $\mathbf{B}\mathbf{A}$.

A general contact transfers both a force impulse $\mathbf{j} = \int \mathbf{f}_c dt$ and a torque impulse $\mathbf{k} = \int \boldsymbol{\tau}_c dt$. Here $\boldsymbol{\tau}_c$ is the free contact moment about the declared contact reference point, excluding the lever-arm moment already represented by $\mathbf{r}_\wedge \mathbf{f}_c$. The complete momentum update is

$$\delta \mathbf{p} = \mathbf{j}, \quad \boxed{\delta \mathbf{L} = \mathbf{r}_\wedge \mathbf{j} + \mathbf{k}}, \quad \delta \mathbf{P} = \mathbf{C}^T \mathbf{j} + \begin{bmatrix} 0 \\ \mathbf{k}/\ell \end{bmatrix}. \quad (8)$$

For a frozen contact lever arm, a distributed traction impulse over a contact patch $\mathcal{A}$ produces

$$\mathbf{j} = \int \int_{\mathcal{A}} \mathbf{t}(\mathbf{x}, t) dA dt, \quad \mathbf{k} = \int \int_{\mathcal{A}} (\mathbf{x} - \mathbf{x}_c) \times \mathbf{t}(\mathbf{x}, t) dA dt. \quad (9)$$

An independently applied couple is included in $\mathbf{k}$ as well. This decomposition accounts for the same physical wrench exactly once. If the reference point shifts by $\mathbf{a}$, then $\mathbf{r}' = \mathbf{r} + \mathbf{a}$ and $\mathbf{k}' = \mathbf{k} - \mathbf{a} \wedge \mathbf{j}$, so $\mathbf{r}' \wedge \mathbf{j} + \mathbf{k}' = \mathbf{r} \wedge \mathbf{j} + \mathbf{k}$.

Choose a fixed contact length $\ell_c > 0$ and define the scaled contact wrench impulse and conjugate contact motion

$$\boldsymbol{\eta} = \begin{bmatrix} \mathbf{j} \\ \mathbf{k}/\ell_c \end{bmatrix}, \quad \mathbf{U}_c = \begin{bmatrix} \mathbf{v}_c \\ \ell_c \boldsymbol{\omega} \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} \mathbf{1} & \mathbf{R} \\ 0 & (\ell_c/\ell) \mathbf{1} \end{bmatrix}. \quad (10)$$

Then the additive form carries the independent torque impulse without a separate update rule:

$$\mathbf{U}_c = \mathbf{K} \mathbf{V}, \quad \delta \mathbf{P} = \mathbf{K}^T \boldsymbol{\eta} = \begin{bmatrix} \mathbf{j} \\ \boldsymbol{\rho} \wedge \mathbf{j} + \mathbf{k}/\ell \end{bmatrix}. \quad (11)$$

Work duality gives

$$\mathbf{V}^T \delta \mathbf{P} = \mathbf{v}_c^T \mathbf{j} + \boldsymbol{\omega}^T \mathbf{k} = \mathbf{U}_c^T \boldsymbol{\eta}. \quad (12)$$

Taking $\ell_c = \ell$ makes the lower-right block of $\mathbf{K}$ an identity. The ideal force-only point-contact specialization sets $\mathbf{k} = 0$ and retains just the three rows $\mathbf{C}$; it cannot represent a finite-patch free torque impulse. Contact wrench/moment models and torsional/rolling friction have established precedents[12, 13].

For planar motion, ω , L , and I are scalars, while translation is two-dimensional. The same construction becomes

$$\mathbf{V} = \begin{bmatrix} v_x \\ v_y \\ \ell \omega \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} p_x \\ p_y \\ L/\ell \end{bmatrix}, \quad \mathbf{M} = \text{diag}(m, m, I/\ell^2), \quad \mathbf{R} = \frac{1}{\ell} \begin{bmatrix} -r_y \\ r_x \end{bmatrix}, \quad \mathbf{C} = [\mathbf{1}_2 \ \mathbf{R}]. \quad (13)$$

Thus $\mathbf{C} \mathbf{V} = \mathbf{v} + \omega(-r_y, r_x)^T$. Include the scalar free torque impulse k through

$$\boldsymbol{\eta} = \begin{bmatrix} j_x \\ j_y \\ k/\ell_c \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} 1 & 0 & -r_y/\ell \\ 0 & 1 & r_x/\ell \\ 0 & 0 & \ell_c/\ell \end{bmatrix}, \quad \Delta L = r_x j_y - r_y j_x + k. \quad (14)$$

The same transpose and duality construction therefore carries over without inventing a planar angular vector.

4 Two bodies and simultaneous contact

Orient a contact normal from body B towards body A. Transfer $+\mathbf{j}, +\mathbf{k}$ to A and $-\mathbf{j}, -\mathbf{k}$ to B, with the force impulse applied at the same world reference point. Use the same contact scale ℓ_c for the two contact maps, even if their body scales differ. The relative contact motion and stacked Jacobian are

$$\mathbf{U} = \begin{bmatrix} \mathbf{u} \\ \ell_c \boldsymbol{\Omega} \end{bmatrix} = \mathbf{K}_A \mathbf{V}_A - \mathbf{K}_B \mathbf{V}_B = \mathbf{G}_w \mathbf{V}, \quad \mathbf{G}_w = [\mathbf{K}_A \ -\mathbf{K}_B], \quad (15)$$

where $\mathbf{u} = \mathbf{v}_{c,A} - \mathbf{v}_{c,B}$ and $\boldsymbol{\Omega} = \boldsymbol{\omega}_A - \boldsymbol{\omega}_B$. For $\mathbf{M} = \text{diag}(\mathbf{M}_A, \mathbf{M}_B)$,

$$\delta \mathbf{P} = \mathbf{G}_w^T \boldsymbol{\eta}, \quad \mathbf{V}^+ = \mathbf{V}^- + \mathbf{M}^{-1} \mathbf{G}_w^T \boldsymbol{\eta}, \quad \mathbf{U}^+ = \mathbf{U}^- + \mathbf{W}_w \boldsymbol{\eta}, \quad (16)$$

with the full contact-wrench inverse effective mass

$$\mathbf{W}_w = \mathbf{G}_w \mathbf{M}^{-1} \mathbf{G}_w^T = \mathbf{K}_A \mathbf{M}_A^{-1} \mathbf{K}_A^T + \mathbf{K}_B \mathbf{M}_B^{-1} \mathbf{K}_B^T. \quad (17)$$

This symmetric positive-semidefinite operator couples force and torque impulses, translation and relative spin, and all contacts sharing a body. A fixed boundary contributes its prescribed contact velocity and angular velocity but no finite inverse mass. Stack every contact's active force and moment components; do not discard off-diagonal coupling in rapid group motion.

For the force-only specialization $\mathbf{k} = 0$, the first three rows of each $\mathbf{K}$ are $\mathbf{C}$, giving $\mathbf{G} = [\mathbf{C}_A \ -\mathbf{C}_B]$ and $\mathbf{W} = \mathbf{G} \mathbf{M}^{-1} \mathbf{G}^T$. The physical predictions are independent of the chosen body and contact coordinate lengths when impulses are transformed consistently.

For a frozen configuration during an impulse, the exact kinetic-energy increment for two dynamic bodies, or one body and a stationary boundary, is

$$\Delta T = \boldsymbol{\eta}^T \mathbf{U}^- + \frac{1}{2} \boldsymbol{\eta}^T \mathbf{W}_w \boldsymbol{\eta}. \quad (18)$$

The linear work term is explicitly $\mathbf{j}^T \mathbf{u}^- + \mathbf{k}^T \boldsymbol{\Omega}^-$. This retains torque-impulse work, including pure relative spin. For a moving prescribed boundary,

$$W_{\text{actuator}} = \mathbf{j}^T \mathbf{v}_b(\mathbf{x}_c) + \mathbf{k}^T \boldsymbol{\omega}_b, \quad \Delta T - W_{\text{actuator}} = \boldsymbol{\eta}^T \mathbf{U}_{\text{relative}}^- + \frac{1}{2} \boldsymbol{\eta}^T \mathbf{W}_w \boldsymbol{\eta}. \quad (19)$$

For multiple contacts, use stacked vectors and the full operator. Free flight still uses the actual rigid-body evolution, changing world inertia and gyroscopic dynamics; the impulsive relation does not replace that evolution.

4.1 Pure spin with a central normal impact

For a sphere on a plane with $\mathbf{r} = -R\mathbf{n}$, a central normal force impulse $\mathbf{j} = j_n \mathbf{n}$ has $\mathbf{r} \wedge \mathbf{j} = 0$. If the sphere spins about $\mathbf{n}$, then $\boldsymbol{\omega} \times \mathbf{r} = 0$ as well. An ideal force-only point contact consequently cannot change this spin. A nonzero torsional contact impulse gives

$$\delta \mathbf{L} = \mathbf{k}, \quad \boldsymbol{\omega}^+ = \boldsymbol{\omega}^- + \mathbf{I}^{-1} \mathbf{k}. \quad (20)$$

For spin perpendicular to $\mathbf{n}$, $\boldsymbol{\omega} \times \mathbf{r}$ generally produces contact slip, so a tangential force impulse can also change spin through $\mathbf{r} \wedge \mathbf{j}_t$. For a non-spherical off-centre contact even a normal impulse may have a nonzero lever-arm moment. The complete wrench formulation includes each mechanism.

Illustrative input	Value
Mass m ; radius R	1 kg; 0.05 m
Inertia I ; lengths ℓ, ℓ_c	0.001 kg m ² ; 0.05, 0.05 m
Incoming normal velocity; axial spin	-1 m/s; 20 rad/s
Prescribed e_n ; free axial impulse k_n	0.6; -0.008 kg m ² /s

The torque impulse is prescribed solely to demonstrate the missing degree of freedom; it is not inferred from a measured outcome or claimed as a material prediction.

Computed quantity	Force impulse only	Force plus torque impulse
Outgoing axial spin, rad/s	20.000	12.000
Outgoing axial angular momentum, kg m ² /s	0.020	0.012
Outgoing normal velocity, m/s	0.600	0.600

The full-wrench energy changes from 0.700 to 0.252 J. The axial angular-momentum change is exactly the prescribed torque impulse, while the central normal force impulse contributes no angular impulse.

5 Normal and tangential restitution

Choose an orthonormal basis $\mathbf{Q} = [\mathbf{n} \ \mathbf{t}_1 \ \mathbf{t}_2]$ and write $\hat{\mathbf{u}} = \mathbf{Q}^T \mathbf{u}$, $\hat{\mathbf{j}} = \mathbf{Q}^T \mathbf{j}$. For an approaching geometrically active impact, the endpoint targets are

$$u_n^+ = -e_n u_n^-, \quad \mathbf{u}_t^+ = -e_t \mathbf{u}_t^-, \quad 0 \leq e_n \leq 1, \quad -1 \leq e_t \leq 1. \quad (21)$$

The signed tangential convention matters: $e_t = -1$ preserves tangential contact velocity, $e_t = 0$ targets sticking, and $e_t > 0$ targets slip reversal. For $-1 < e_t < 0$, tangential contact motion continues in its original direction with reduced magnitude. For example, $e_t = -.2$ gives $\mathbf{u}_t^+ = .2\mathbf{u}_t^-$: this sign describes the direction of a velocity ratio, not negative elastic stiffness or energy creation. Normal and tangential restitution describe different responses and need not be equal. These are *contact-point* velocities; a centre-of-mass tangential speed ratio is not generally e_t . In coupled geometries, actual total energy must still be checked.

Set $\mathbf{E} = \text{diag}(e_n, e_t, e_t)$. In the force-only specialization $\mathbf{k} = 0$, if the isolated full-rank contact can realize all targets, its impulse solves

$$\widehat{\mathbf{W}}\hat{\mathbf{j}} + (\mathbf{1} + \mathbf{E})\hat{\mathbf{u}}^- = 0, \quad \widehat{\mathbf{W}} = \mathbf{Q}^T \mathbf{W} \mathbf{Q}. \quad (22)$$

With an independent torque impulse the translational endpoint instead reads

$$\widehat{\mathbf{W}}_f \hat{\mathbf{j}} + \widehat{\mathbf{W}}_{fk} \hat{\mathbf{k}} / \ell_c + (\mathbf{1} + \mathbf{E})\hat{\mathbf{u}}^- = 0. \quad (23)$$

Here the blocks are taken from $\mathbf{W}_w$ after rotating both its force and moment components into the contact basis. Thus the three translational restitution targets do not determine a six-component contact wrench. Its moment components require a separately specified physical contact law or measured torque impulse. Any rotational restitution targets must likewise be independently justified, with actual coupled energy and friction limits checked.

This residual form keeps addition visible without concealing physical signs. In a contact network, unilateral conditions, friction capacity, and coupled energy constraints may prevent all endpoint targets being realized simultaneously. Bounds on individual coefficients alone do not establish passivity of the coupled system; this issue has prior analysis, including the Moreau-law discussion by Acary and Collins-Craft[11]. Test the actual energy increment using physical incoming velocities, not a restitution-shifted solver residual.

6 Friction and the distinction between capacity and compliance

The circular Coulomb impulse capacity is

$$j_n \geq 0, \quad \|\mathbf{j}_t\| \leq \mu j_n. \quad (24)$$

The current endpoint model seeks its tangential restitution target when capacity permits and limits the impulse when it does not. This capacity condition is not by itself a complete time-resolved friction law. In particular, a positive e_t can represent stored shear energy returning during impact; a rigid, purely dissipative sliding law alone does not explain that return.

For a sphere with radius R , inertia $I = \alpha m R^2$, zero initial spin, and fixed plane, the uncapped tangential impulse is

$$\frac{j_t}{m} = -\frac{\alpha}{1 + \alpha} (1 + e_t) u_t^-, \quad |\omega^+| = \frac{(1 + e_t) |u_t^-|}{(1 + \alpha) R}. \quad (25)$$

These sphere formulas assume $\mathbf{k} = 0$. When required capacity is exceeded, clamp this scalar impulse to $[-\mu j_n, \mu j_n]$; normal and tangential targets are uncoupled only in this special spherical geometry. Actual non-spherical inertia and lever arms produce coupling through $\mathbf{W}$.

A physically motivated extension would evolve normal indentation and tangential elastic displacement over a finite contact patch, integrate its force and moment, and account for elastic energy storage plus nonnegative dissipation. Restitution would then be an output or a constrained calibration target; adding endpoint restitution on top of an already dissipative compliant rebound risks double counting. A measured distributed moment is distinct from arbitrarily shifting a contact point until one outcome matches.

6.1 Moment capacity is part of the contact law

The translational Coulomb inequality alone does not specify $\mathbf{k}$. A finite patch can supply torsional and rolling moments, or moments from a distributed normal load. For example, simplified necessary capacity bounds may be written

$$|k_n| \leq b_{\text{twist}} j_n, \quad \|\mathbf{k}_t\| \leq b_{\text{roll}} j_n, \quad (26)$$

where $k_n = \mathbf{n}^T \mathbf{k}$ and $\mathbf{k}_t = \mathbf{k} - k_n \mathbf{n}$. The physical coefficients $b_{\text{twist}}, b_{\text{roll}}$ have units of length; they are distinct from the freely chosen coordinate lengths ℓ, ℓ_c . Their values and applicability must come from a measured or derived patch law. Separate force/moment bounds are generally not a complete jointly admissible patch-wrench law; integrating common tractions can couple the capacities. No new moment coefficient is fitted to the existing comparison outcomes.

The native comparisons retained here use force-only contacts. Their contact forces can change spin through lever-arm torque, but a separate free contact moment is absent. They therefore do not validate pure torsional spin resistance or a complete contact-wrench law. The corrected derivation identifies that missing mechanism explicitly; it does not silently change the historical comparison numbers.

7 Efficient practical matrix calculations

The scaled notation does not require constructing a dense global inverse. Store each body’s mass, shape-frame inertia, attitude, and fixed reference length. For attitude matrix $\mathbf{A}$, rotate inertia once per pose: $\mathbf{I}_{\text{world}} = \mathbf{A} \mathbf{I}_{\text{shape}} \mathbf{A}^T$. Apply the inverse scaled mass to a vector by dividing its translation block by m and solving a three-by-three inertia system for its rotation block, multiplied by ℓ^2 . In 2D the latter is scalar division by I/ℓ^2 . Use factorizations or the rotated inverse of a fixed shape inertia; avoid numerical inversion of a dense many-body matrix.

For a trial stacked wrench impulse $\boldsymbol{\eta}$, the matrix-free action $\mathbf{W}_w \boldsymbol{\eta}$ has three steps:

1. Accumulate each incident signed force and torque impulse through $\mathbf{K}_i^T \boldsymbol{\eta}_i$ into each body’s scaled momentum increment.
2. Apply the body’s inverse mass block once to its accumulated increment.
3. Gather the resulting signed contact velocity and scaled relative-spin increments through each $\mathbf{K}_i$.

Shared bodies automatically generate the off-diagonal contact coupling. Coupled matrix-free contact evaluation is already taught in the SIGGRAPH 2022 contact tutorial[10]. This operation scales with bodies and contacts rather than the square of the global contact count. Assemble small island matrices when useful for factorization or diagnostics; preserve the full coupled law and acceptance gates. Independent pairwise impulses are not an equivalent substitute.

A practical prediction requires incoming translation and spin, mass and inertia, contact geometry and normal, and explicit e_n, e_t, μ . For each active approaching contact, construct the restitution target and solve the simultaneous system with nonnegative normal impulse and the exact circular tangential capacity. Report whether each tangential target was realized or capacity-limited, outgoing velocities and spin, momentum balance, and work-corrected energy.

Use the physical incoming contact velocity in the energy check. The native implementation has validated coefficient bounds, but not every large-scene trajectory refinement is qualified.

The accompanying `matrix.tools.py` demonstrates scaled variables and matrix-free contact operations without changing production code. Its three-dimensional operations pass 100 seeded four-body, three-contact shared-chain comparisons between dense and matrix-free Delassus actions (maximum discrepancy 2.84×10^{-14}). Its native 2D and 3D prediction wrappers reproduce their retained demonstration trajectories exactly (`prediction-examples.json`). The planar restitution build remains isolated and has not qualified the full benchmark baseline. Each wrapper passes e_n, e_t, μ explicitly to the actual solver; the 3D illustrative pair assigns the square root of the pair coefficient to each body because Bullet multiplies body friction, while the 2D wrapper uses the Box2D square-root mixing convention. These checks do not establish a speedup or replace native simultaneous-solver verification. The demonstrated native force-only material/contact inputs are e_n, e_t, μ . The full wrench matrices derived above are evaluated separately by `wrench_impulse_audit.py`: 100 random dynamic 3D pairs and 100 planar controls check direct force-plus-couple updates, reference-point shifts, coordinate scaling, angular balance and torque-inclusive energy/boundary work. These controls prescribe $\mathbf{k}$; they do not implement or validate a predictive native moment law. Rolling resistance, torsional resistance, independent contact moments, and compliant patch stiffness/damping would require additional measured laws and validation. They are not silently assumed present merely because their symbols could be added to a matrix.

8 Worked three-dimensional corner impact

A rotated cuboid striking a stationary plane makes the full 3D coupling visible: its world inertia is non-diagonal, its contact is off centre, and both incoming translation and spin are three-component vectors. The frozen one-contact calculation below evaluates the force-only specialization $\mathbf{k} = 0$ of the matrix equations above.

Illustrative input	Value
Mass m	2 kg
Half extents (a, b, c)	(.20, .15, .10) m
Reference length ℓ	.20 m (coordinate choice)
Incoming $\mathbf{v}^-$	(1, -.6, -2) m/s
Incoming $\boldsymbol{\omega}^-$	(2, -1, 3) rad/s
Normal $\mathbf{n}$; tangents $\mathbf{t}_1, \mathbf{t}_2$	$\mathbf{e}_z; \mathbf{e}_x, \mathbf{e}_y$
Normal restitution e_n	.6
Tangential restitution e_t	.3
Pair friction μ	.6
Boundary	Stationary plane, $z = 0$

These are prescribed illustration inputs, not a published material profile or fitted experimental values. The matrix equations below retain symbolic material coefficients. Attitude and lever-arm signs are fixed by the actual box geometry.

$$\mathbf{A} = \frac{1}{9} \begin{bmatrix} 1 & 4 & 8 \\ 4 & 7 & -4 \\ -8 & 4 & -1 \end{bmatrix}, \quad \mathbf{r}_{\text{shape}} = \begin{bmatrix} a \\ -b \\ c \end{bmatrix}, \quad \mathbf{r} = \mathbf{A} \mathbf{r}_{\text{shape}}. \quad (27)$$

The supporting vertex is unique: every other vertex lies above the plane when the centre of mass is placed at $-\mathbf{r}$. For a homogeneous cuboid,

$$\mathbf{I}_{\text{shape}} = \frac{m}{3} \text{diag}(b^2 + c^2, a^2 + c^2, a^2 + b^2), \quad \mathbf{I} = \mathbf{A} \mathbf{I}_{\text{shape}} \mathbf{A}^T. \quad (28)$$

$$\mathbf{r} = \begin{bmatrix} 0.044444 \\ -0.072222 \\ -0.255556 \end{bmatrix} \text{ m}, \quad \mathbf{I} = \begin{bmatrix} 0.039774 & -0.003868 & 0.000329 \\ -0.003868 & 0.032675 & 0.005021 \\ 0.000329 & 0.005021 & 0.024218 \end{bmatrix} \text{ kg m}^2. \quad (29)$$

All three spin components are retained. The six-component state, mass block and three-by-six contact map are

$$\mathbf{V}^- = \begin{bmatrix} \mathbf{v}^- \\ \ell \boldsymbol{\omega}^- \end{bmatrix}, \quad \mathbf{M} = \begin{bmatrix} m \mathbf{1}_3 & 0 \\ 0 & \mathbf{I}/\ell^2 \end{bmatrix}, \quad \mathbf{C} = [\mathbf{1}_3 \ \mathbf{R}]. \quad (30)$$

$$\mathbf{R} = \begin{bmatrix} -0.000000 & -1.277778 & 0.361111 \\ 1.277778 & -0.000000 & 0.222222 \\ -0.361111 & -0.222222 & -0.000000 \end{bmatrix}, \quad \mathbf{u}^- = \mathbf{C} \mathbf{V}^-. \quad (31)$$

The stationary plane contributes zero inverse mass. Use the full tensor, then rotate into the ordered contact basis (n, t_1, t_2) :

$$\mathbf{W} = \mathbf{C} \mathbf{M}^{-1} \mathbf{C}^T, \quad \widehat{\mathbf{W}} = \mathbf{Q}^T \mathbf{W} \mathbf{Q}, \quad \mathbf{Q} = [\mathbf{e}_z \ \mathbf{e}_x \ \mathbf{e}_y]. \quad (32)$$

$$\widehat{\mathbf{W}} = \begin{bmatrix} 0.716353 & 0.448661 & -0.489459 \\ 0.448661 & 3.059430 & -0.012536 \\ -0.489459 & -0.012536 & 2.224217 \end{bmatrix} \text{ kg}^{-1}. \quad (33)$$

Its off-diagonal entries couple normal and tangential impulses: solving only its diagonal would give a different collision. Solve the small symmetric system rather than form an explicit inverse:

$$\widehat{\mathbf{W}} \widehat{\mathbf{j}} + (\mathbf{1} + \mathbf{E}) \mathbf{Q}^T \mathbf{u}^- = 0, \quad \mathbf{E} = \text{diag}(e_n, e_t, e_t), \quad (34)$$

$$\mathbf{j} = \mathbf{Q} \widehat{\mathbf{j}}, \quad (35)$$

$$\mathbf{V}^+ = \mathbf{V}^- + \mathbf{M}^{-1} \mathbf{C}^T \mathbf{j}. \quad (36)$$

Check $j_n \geq 0$, $\|\mathbf{j}_t\| \leq \mu j_n$ and the actual kinetic-energy increment before accepting that endpoint solution. Here the unconstrained target is admissible, so no friction-capacity adjustment is required.

Computed result	Value
$\mathbf{u}^-$, world (x, y, z)	(1.472222, 0.044444, -2.100000) m/s
$\widehat{\mathbf{j}}$, contact (n, t_1, t_2)	(6.670214, -1.597877, 1.432858) kg m/s
$\mathbf{v}^+$	(0.201062, 0.116429, 1.335107) m/s
$\boldsymbol{\omega}^+$	(-0.535728, 2.560464, 0.160786) rad/s
$\mathbf{u}^+$, world (x, y, z)	(-0.441667, -0.013333, 1.260000) m/s
$\ \mathbf{j}_t\ /j_n$	0.321763
$T^-; T^+$	5.559516; 1.956965 J

The example changes translation and all three spin components while satisfying both restitution targets. Energy decreases; the stationary plane receives the opposing impulse. Linear and angular momentum are checked through their impulse balances, rather than assuming conservation for the cuboid alone.

The retained calculation compares scaled matrix operations with independent unscaled cross-product updates, checks the impulse/energy identities and the supporting vertex, and repeats the solve at four reference lengths and three rotated tangent bases. Physical impulses and outgoing states remain unchanged to the reported numerical precision. These are algebra and geometry checks, not an experimental accuracy claim or a native time-step replay.

9 Mapping prediction deviations

For each measured comparison, collect a magnitude signature

$$\mathbf{q} = \begin{bmatrix} |v_n| \\ |v_t| \\ \ell |\boldsymbol{\omega}| \end{bmatrix}, \quad \phi = \arccos \frac{\mathbf{q}_{\text{measured}}^T \mathbf{q}_{\text{predicted}}}{\|\mathbf{q}_{\text{measured}}\| \|\mathbf{q}_{\text{predicted}}\|}, \quad \varepsilon = \frac{\|\mathbf{q}_{\text{predicted}}\|}{\|\mathbf{q}_{\text{measured}}\|} - 1. \quad (37)$$

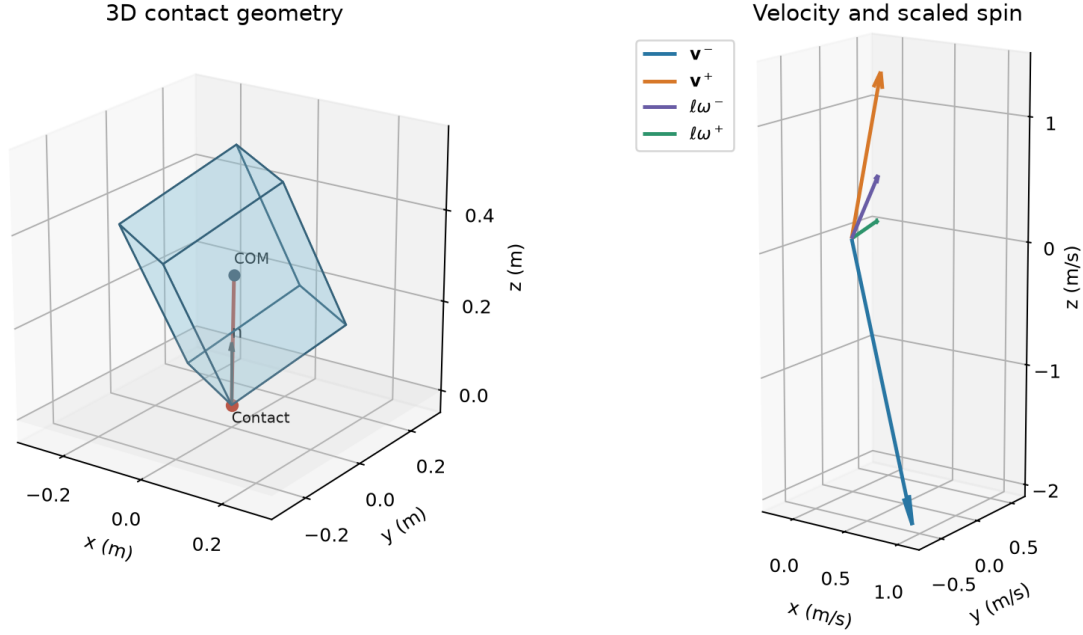


Figure 1: The rotated cuboid contacts the plane at its unique lowest corner. The right panel compares incoming and outgoing translation and length-scaled spin in the same velocity units; arrows begin at a common origin for comparison. The impulse acts at the corner, while spin is defined about the centre of mass.

Each case becomes one point (ϕ, ε) : exact agreement is at the origin. The horizontal axis shows an incorrect mix of translation and rotation; the vertical axis shows too much or too little combined motion. It is the angle between observable signatures, not a spatial heading error. Colours identify collision types or declared contact regimes. Inputs need not be artificially separated, and different impacts can naturally share the same error coordinates.

The plotting policy takes ℓ equal to the reported radius, without fitting it to the errors. This converts spin into peripheral-speed units. Unlike physical predictions, numerical angles and norms of coordinate signatures depend on the chosen plotting metric. Only reported or explicitly reconstructed observables are used; unavailable vector signs and orientation are not invented. The main comparison below uses documented inputs; the historical fit-based map is separated in the appendix.

10 Selecting documented material-pair inputs

The active modeling policy uses published experimental parameter profiles and performs no project fitting of material or extra contact parameters. Each profile identifies the material pair, specimen/surface condition, collision geometry and reported size. Missing coefficients remain unavailable rather than being supplied by a fit. Uncertainties and source validity conditions are retained; a generic material name does not define a universally exact collision constant.

The separate catalog contains eleven complete published triples. The following subset illustrates lookup inputs; equations above remain symbolic. Cornell’s chart describes small spheres, while Joseph and Hunt document the air/Zerodur profiles over a specified normal-speed range[1, 2].

Documented pair	Diameter mm	e_n	e_t	μ
Soda glass / same	3.18	$.97 \pm .01$	$.44 \pm .07$	$.092 \pm .006$
Fresh glass / same	2.97	$.922 \pm .021$	$.37 \pm .07$	$.048 \pm .006$
Spent glass / same	2.97	$.972 \pm .015$	$.25 \pm .08$	$.177 \pm .020$
Soda glass / smooth Al	3.18	$.831 \pm .009$	$.31 \pm .06$	$.125 \pm .007$
Steel / Zerodur, air	12.7	$.97 \pm .02$	$.34 \pm .07$	$.11 \pm .003$
Glass / Zerodur, air	12.7	$.97 \pm .02$	$.39 \pm .06$	$.106 \pm .008$

The catalog is in `research/documented-materials/catalog.json`. The pair loader realizes a published pair friction value through the backend’s actual combination rule; it does not reinterpret pair friction as a universal single-body property. Eighty-one native analytic/energy controls pass with source coefficients unchanged. These are integration checks, not measured-motion validation.

For the original Cornell glass-binary worksheet, twenty-four source-reconstructed impacts were predicted with the fixed published profile and no project coefficient optimization. Outgoing relative normal-velocity RMSE is 0.03667 m/s and COM tangential-velocity RMSE is 0.01780 m/s. The parameter chart and worksheet may share characterization trials, so this is reproduction rather than independent cross-dataset validation. Spin is inferred from the author’s angular-momentum reconstruction under sphere inertia, not independently measured. The plate worksheet has a different parameter summary and is not silently merged into this comparison.

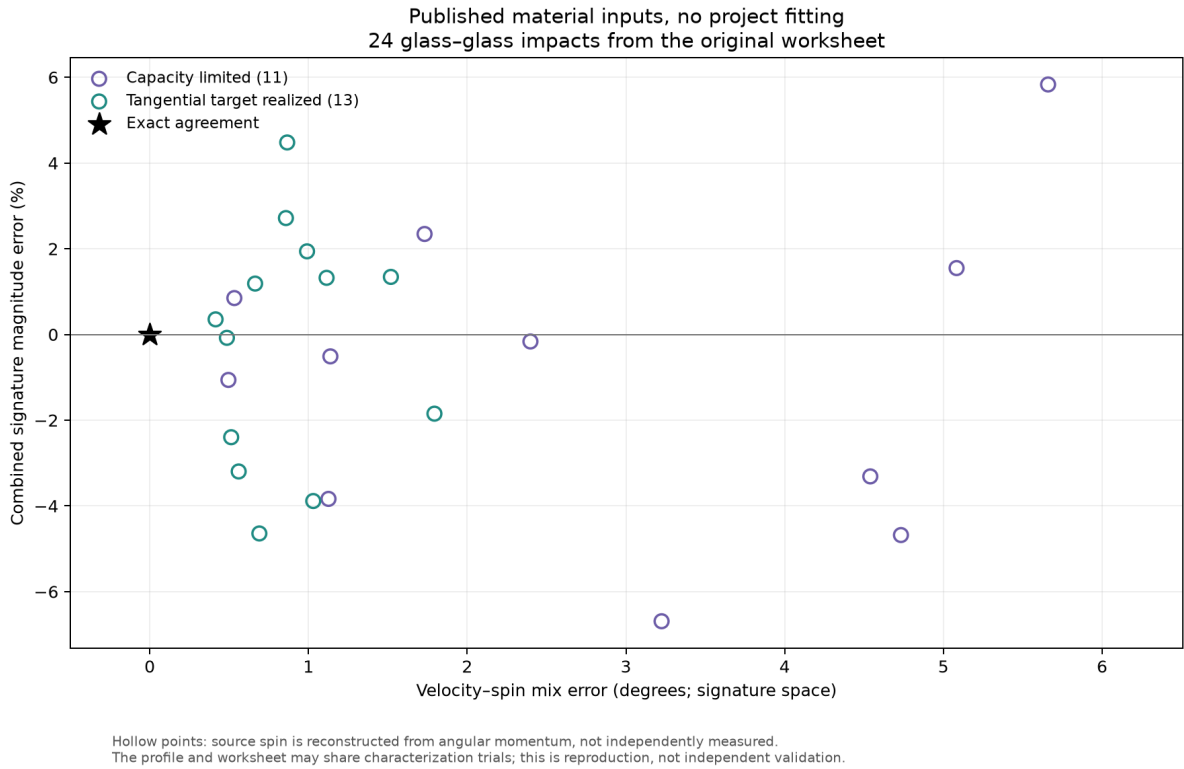


Figure 2: Source-backed inputs without project fitting: the same fixed glass/glass profile predicts twenty-four incoming states. Hollow points identify that spin is reconstructed rather than independently measured. Colour distinguishes the model’s tangential-capacity regimes; it is not a classification of measured force history. The existing radius policy and observation values are unchanged.

For limestone/concrete, a compatible complete contact restitution/friction profile has not been verified. The old fitted rock coefficients below are historical diagnostics and are excluded from material lookup. Likewise, the rubber comparison’s assumed friction is not a documented pair value. Their dots cannot be relabeled as independently specified material predictions.

11 What would establish a stronger model?

First replace approximate shapes, inertias, and initial states with independently measured quantities wherever available. For rubber, compare a time-resolved compliant patch against the point model using independently characterized laws and parameters fixed before held-out impacts. Evaluate outgoing translation and rotation together, contact duration where measured, and a complete energy ledger. Translational energy alone need not be conserved: rotation, deformation, support motion, and dissipation exchange energy. Do not fit a separate force offset for every target outcome and call that an independent prediction.

For non-spherical rocks, contact location already changes the Jacobian and effective mass without changing restitution. Introduce a material map only if repeated experiments with measured geometry and state show a reproducible residual dependence that transfers to unseen contacts. The reference length ℓ must never serve as an adjustable mechanism for improving agreement: it changes notation, not physics.

A Archived conditional comparisons and superseded fits

This appendix preserves earlier calculations for review. No new fitting is performed, and its rock coefficients are excluded from the material catalog and current prediction policy. Negative tangential restitution is permitted by the sign convention, but the sign alone does not justify interpreting an approximate-shape fit as a material property.

Ball	Surface	e_n	e_t
Superball	granite	0.78	0.49
Superball	rubber	0.78	0.41
Superball	Superball pad	0.78	0.57
Superball	tennis strings	0.91	-0.10
golf	granite	0.87	0.03
golf	rubber	0.59	0.51
golf	Superball pad	0.81	0.55
golf	tennis strings	0.91	-0.01

These surface restitution inputs are supplied from the target experiments; the table is separate from the prediction-error tables.

The following numbers describe the frozen point-contact model, not a newly validated compliant model. The public rubber experiments report the same 58 mm, 103 g Superball on four surfaces [3]. The incoming angle is approximately 25 ± 1 degrees to the vertical. The evaluation uses the separate assumption table below; capacity is inactive. All derivations retain symbolic material inputs. Both restitution coefficients come from the target experiment. The remaining comparison is spin factor $S = |\omega^+|/v_{\text{in}}$ in rad/m.

Surface	Measured S	Model S	Overlap?
Granite	$14.9 \pm .1$	15.510	Yes
Rubber sheet	$14.5 \pm .1$	14.677	Yes
Superball pad	$18.2 \pm .1$	16.343	No
Tennis strings	$9.0 \pm .1$	9.368	Yes

“Overlap” includes input-angle and reported $e_t \pm .01$ uncertainty; it does not mean each central estimate lies in the narrow measured-spin interval. The respective prediction ranges are [14.827, 16.196], [14.025, 15.332], [15.628, 17.060], and [8.916, 9.826]. No single adjusted sphere inertia factor explains all four within these intervals. These checks do not independently validate friction, because μ is a hypothesis and the supplied restitution values were measured in the very impacts being compared.

The same publication also gives a 42.8 mm, 45 g golf ball. This is a different material and construction, not a controlled change of Superball size. Under the same conditional homogeneous-inertia calculation, the additional rows are:

Golf-ball surface	Measured S	Model S	Overlap?
granite	$14.8 \pm .1$	14.529	Yes
rubber	$20.5 \pm .1$	21.300	Yes
Superball pad	$23.6 \pm .1$	21.864	No
tennis strings	$15.0 \pm .1$	13.965	No

These golf-ball predictions are analytic conditional comparisons, not additional native runs. Their inertia is assumed rather than independently measured; neither inferred contact-force offsets nor extra moments are applied to the predictions.

Input set	e_n	e_t	μ	α	Status
Rubber/golf	Surface rows	Surface rows	.9	.4	Restitution supplied from target experiments; friction and homogeneous inertia assumed.
Limestone fixed fit	.492986	-.191582	.811484	.4	Fitted restitution/friction; approximate sphere inertia.
Synthetic geometry	.6	.6	.2 or 2	–	Prescribed controls; actual shape inertia used.

This input table is kept separate from the symbolic calculation. Numerical evaluation draws its inputs from the table; no experimental material value is substituted into the derivation. The homogeneous inertia assumption for the golf ball is particularly conditional because its construction differs from the Superball.

The limestone dataset contains 75 impacts of approximately 10 and 20 cm faceted rocks on concrete [4]. The earlier study trained on 50 impacts at release heights 2.5 and 3.5 m and withheld all 25 at 4.5 m. Actual meshes, individual inertia tensors, contact attitudes, and signed initial-spin vectors are unavailable. The comparison therefore uses ideal spheres and zero incoming spin; the paper bounds initial spin magnitude by approximately 3 rad/s. The fitted coefficients are listed in the separate input table above; they are effective fitted values, not independently measured material properties. In particular, the negative tangential coefficient could compensate for missing geometry, inertia or initial spin. It is not promoted to a limestone/concrete property.

Model	Joint error dimensionless	v_n RMSE m/s	v_t RMSE m/s	ω RMSE rad/s
Fixed coefficients	.121084	1.0044	1.3047	12.8765
Separate per size	.119360	.9958	1.2755	11.2461
Angle-dependent	.109303	.7008	1.2673	12.7285

Joint error is the root mean square over all three residual components, with angular residual multiplied by radius and each impact normalized by incoming speed. The angle-dependent candidate improves this split by 9.7%, but reverses its advantage when one entire plate-angle group is withheld: pooled joint errors are .119679 for fixed, .120366 for size-dependent, and .130625 for angle-dependent coefficients. Thus increased parameter freedom does not establish a transferable contact-location or direction-dependent material law.

Native implementation checks are separate from experimental accuracy: all 75 held-out analytic-to-native predictions (25 impacts times three models) agree within 3.13×10^{-13} in their checked endpoint components and pass the energy check. Ninety-six synthetic non-spherical geometry cases also pass with the separate control parameters above and two prescribed friction levels. A separate 200-case seeded two- and three-dimensional coordinate audit (retained in `notation-audit.json`) confirms kinetic energy, impulse duality, torque, and Delassus invariance over reference lengths from 10^{-5} to 10^3 m. Those checks verify signs, geometry, capacity, and implementation; they do not prove the faceted experimental rocks were replayed exactly.

A separate recent finite-contact candidate was rejected: fourfold held-out spin RMSE worsened from 0.998608 to 1.235076 despite passing its 36 analytic and 40 differential-equation checks. It was not promoted into the production model. Passing mathematical controls is necessary but insufficient evidence of improved reality prediction.

Public dataset	Retained evidence	Present comparison limits
Cross rub- ber/golf	Eight ball-surface rows	Conditional spin predictions; target-supplied restitution and assumed inertia.
Wang lime- stone	75 impacts; 25 held out	Approximate spheres; actual facets, tensor and contact attitude missing.
Tschamut 2014	2219 before/after scalar rotation records across 74 tests	Rock masses joined; scalar spin cannot recover signed angular vectors, contact normals or exact forward replay.
Chant Sura	82 reconstruction CSV files; 41 with finite gyro and rotational-energy data	Manufactured concrete shapes; vector motion available, but attitude/frame and angular-unit interpretation require verification.

The latter two inventories identify useful future evidence, not additional completed validation. Their primary landing pages are <https://www.envidat.ch/metadata/tschamut2014> and

<https://doi.org/10.16904/envidat.174>. Scalar angular speed ratios must not be substituted for signed contact tangential restitution. Source and licensing details are retained with the corresponding repository inventories.

B Archived deviation map

This figure uses the metric in Section 9, with the original unfitted radius policy. It is retained as historical evidence and is not a comparison using a complete independently documented material-pair profile.

The map retains 25 held-out limestone impacts under the fixed-coefficient model and eight ball-surface summary observations. Only reported magnitudes are used; unavailable angular-vector signs and orientation are not invented.

For the ball rows, the tangential speed is reconstructed from the published relation $v_t = R\omega - e_t v_{in,t}$, and the normal speed uses the target-supplied e_n . Those are conditional signatures, not independent full-state measurements. Incoming speed cancels from both map coordinates. The thin ball error lines summarize variation over reported incidence, restitution and spin uncertainty; they are not confidence intervals. Rock uncertainty is not available. Synthetic implementation checks are excluded from this reality diagram.

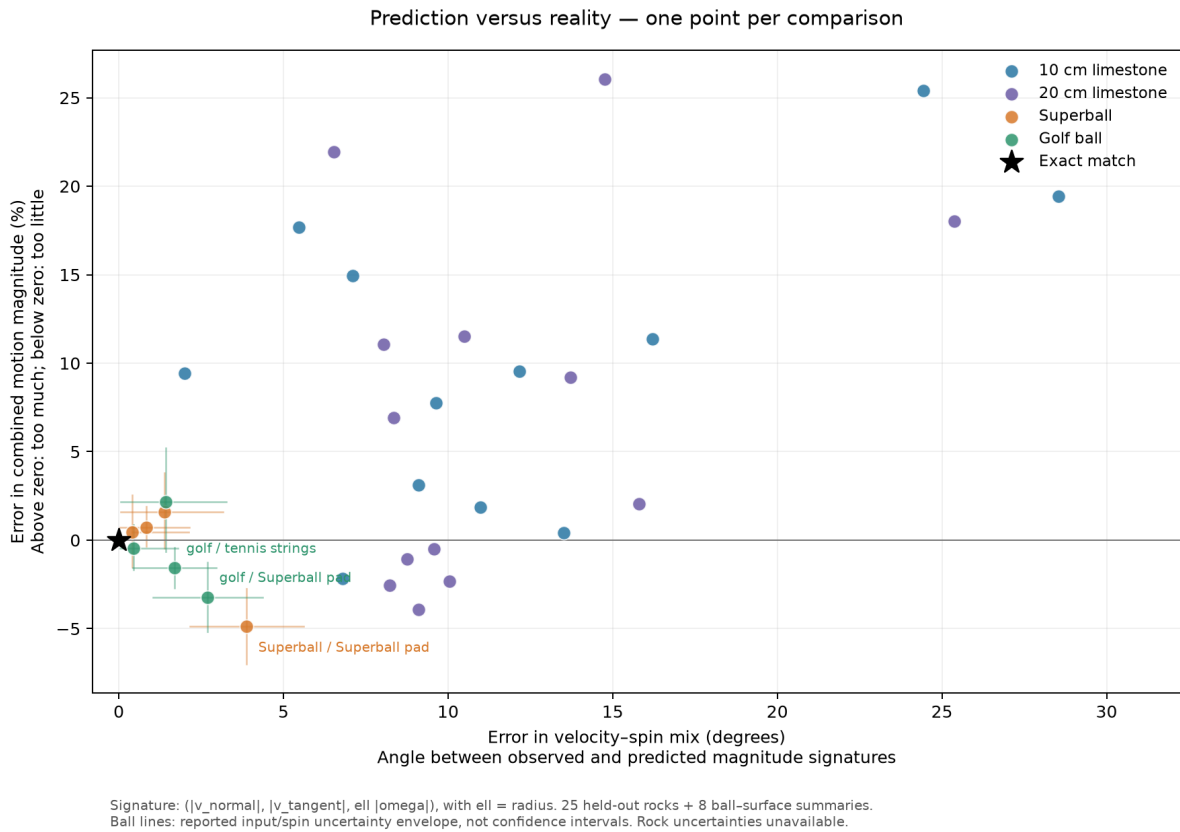


Figure 3: Historical conditional and fit-based deviation map. A dot near the origin agrees in combined magnitude and the relative mix of translation and spin. The map is a two-coordinate summary, not a full signed state comparison. The accompanying standalone `deviation-map.html` supports point inspection; `deviation-map.json` retains the observable vectors and mapping assumptions.

References

- [1] Cornell Grain Flow Research, impact parameter chart and experimental method. <https://grainflowresearch.mae.cornell.edu/impact/data/Impact%20Results.html>; <https://grainflowresearch.mae.cornell.edu/impact/impact.html>.
- [2] G. G. Joseph and M. L. Hunt, *Oblique particle-wall collisions in a liquid*, J. Fluid Mech. 510 (2004), 71–93. Dry profiles on p78. <https://doi.org/10.1017/S002211200400919X>.
- [3] R. Cross, *Enhancing the Bounce of a Ball* (2010), Table I. Author-hosted manuscript: <https://physics.usyd.edu.au/~cross/PUBLICATIONS/48.%20EnhanceBounce.pdf>.
- [4] Wang et al., *Effects of the impact angle on the coefficient of restitution in rockfall*, Natural Hazards and Earth System Sciences 18 (2018), 3045–3060. <https://doi.org/10.5194/nhess-18-3045-2018>. Publisher supplementary spreadsheet provides the impact observations.
- [5] R. Featherstone, *General Comments on Coordinate Transforms*, Spatial Vector and Rigid-Body Dynamics documentation. Motion/force duality and inertia transformation. <https://royfeatherstone.org/spatial/v2/xforms.html>.
- [6] K. M. Lynch and F. C. Park, *Modern Robotics*, section 3.4, Wrenches. Twist–wrench power pairing. <https://modernrobotics.northwestern.edu/nu-gm-book-resource/3-4-wrenches/>.
- [7] J. Angeles, *On the Optimum Dimensioning of Robotic Manipulators*, author-hosted manuscript, sections 1–2. Characteristic-length normalization and homogeneous Jacobians. <https://cim.mcgill.ca/~rmsl/Index/Documents/MECH573/robotdes0312.pdf>.
- [8] T. H. Vose, P. Umbanhowar and K. M. Lynch, *Sliding Manipulation of Rigid Bodies on a Controlled 6-DoF Plate*, Robotics: Science and Systems VII (2011). Appendix footnote 1, PDF p7. <https://www.roboticsproceedings.org/rss07/p43.pdf>.
- [9] Y. Zhang, H.-J. Su and Q. Liao, *Mobility criteria of compliant mechanisms based on decomposition of compliance matrices*, Mechanism and Machine Theory 79 (2014), 80–93. Section 3.3, equation 21. <https://doi.org/10.1016/j.mechmachtheory.2014.04.010>.
- [10] *Contact and Friction Simulation for Computer Graphics*, SIGGRAPH 2022 course notes. Matrix-free tutorial, PDF pp163–164. https://siggraphcontact.github.io/assets/files/SIGGRAPH22_friction_contact_notes.pdf.
- [11] V. Acary and N. A. Collins-Craft, *On the Moreau–Jean scheme with the Frémond impact law: energy conservation and dissipation properties for elastodynamics with contact, impact and friction*, JTCAM (2025). Section 2.5. <https://doi.org/10.46298/jtcam.13480>.
- [12] Drake documentation, *Hydroelastic Contact User Guide*. Distributed contact force and torque. https://drake.mit.edu/doxygen_cxx/group__hydroelastic__user__guide.html.
- [13] MuJoCo documentation, *Computation: Contact*. Torsional and rolling friction and higher-dimensional contact wrenches. <https://mujoco.readthedocs.io/en/stable/computation.html>.